Oracle Dynamic Scaling Engine

Scale Up/Down Automation Utility in OCI EXAC@C /EXADB-D/Multicloud

Office of CTO | EMEA Technology Engineering

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Agenda

1. ExaCC & ExaCS CPU SCALING OVERVIEW
    1. MANUAL OCPUS Scaling
2. Oracle Dynamic Scaling
    1. Automate OCPUs Scaling
3. Dynamic CPU Count

OCPU Scaling OverView

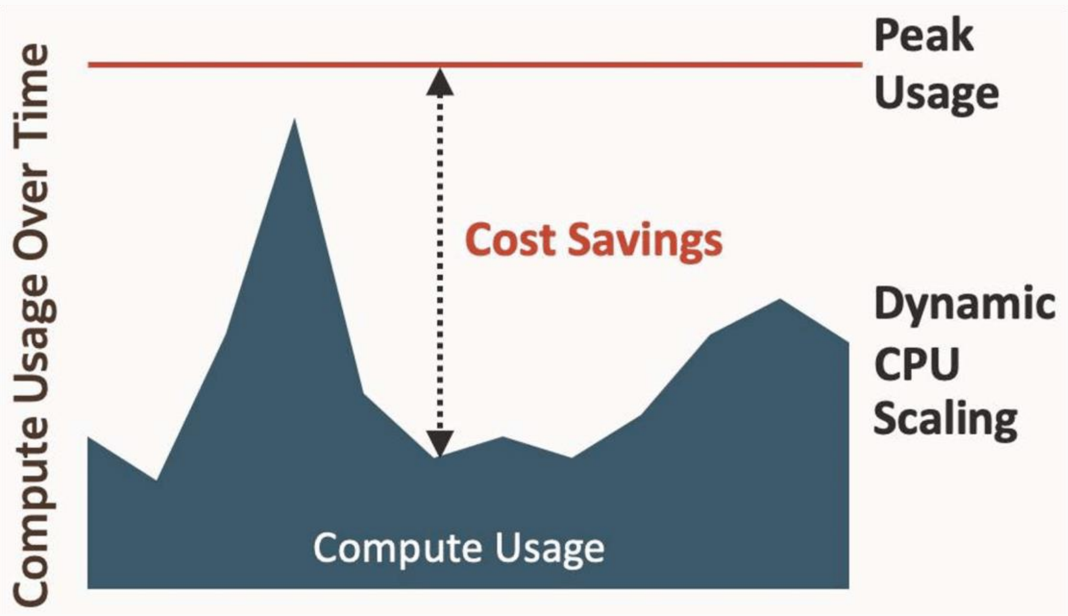
DYNAMIC SCALING

A common advantage between Exadata Database Service (ExaDB) and Oracle Autonomous Database (ADB) is the elasticity of CPU and storage. Just consider the importance of vertically scaling and virtual machines.

ExaDB database licensing costs are based on the number of peak OCPUs, or cores, in use. For ADB, it's based on the number of peak ECPUs. Scaling the cores in use by VMs as the workload scales up and down is important to reducing costs and is one of the greatest advantages of cloud over on-premises deployments.

- Size for average/baseline, not peak usage
- Pay for actual compute usage
- Scale based on workload demand
- Online scaling with ExaDB and ADB

With both ExaDB and ADB deployment, if customers have a spike in demand for their workloads, they can scale up OCPU/ECPU consumption to accommodate it. Then, when demand drops, they can immediately scale down consumption to reduce costs. Unlike most cloud database services, ExaDB and ADB allow you to scale online, with no disruption to the database service.



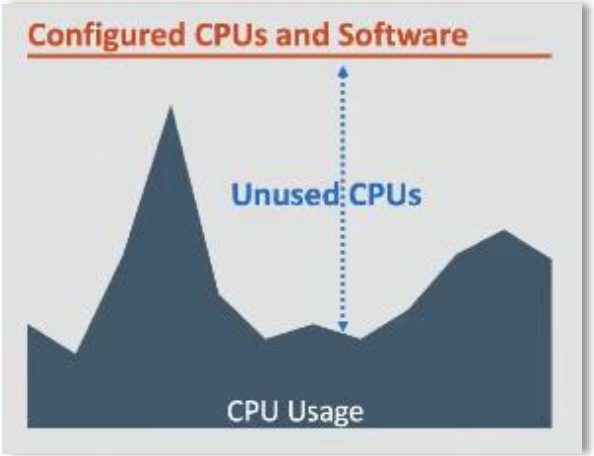
Elasticity in Exadata Database Service (ExaDB) and Autonomous Database (ADB)

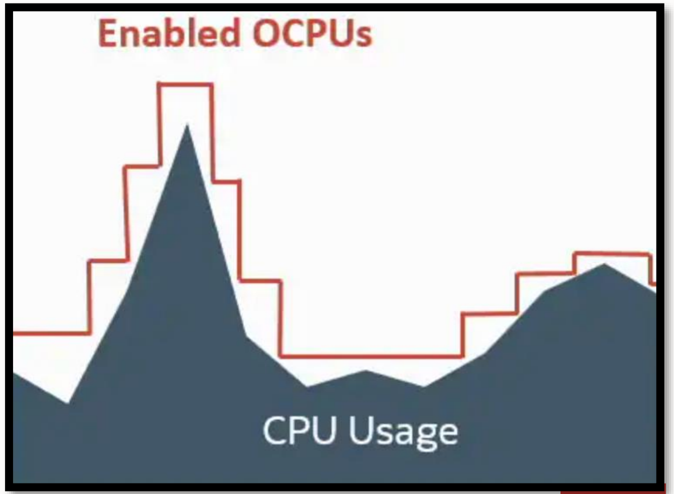
Key Advantage: Vertical Scaling with Virtual Machines

- ExaDB Licensing Costs: Based on peak OCPUs (cores) in use.
- ADB Licensing Costs: Based on peak ECPUs in use.
- Cost Optimization: Scale CPU and storage up or down with workload demand, reducing costs compared to on-premises deployments.

Why Elasticity Matters

- Size for Baseline, Not Peak Usage: Avoid over-provisioning by scaling resources dynamically.
- Pay for Actual Usage: Only pay for the compute and storage you use.
- Scale Based on Demand: Adjust resources in real-time to match workload requirements.
- Online Scaling: Scale OCPUs/ECPUs without disrupting database operations.



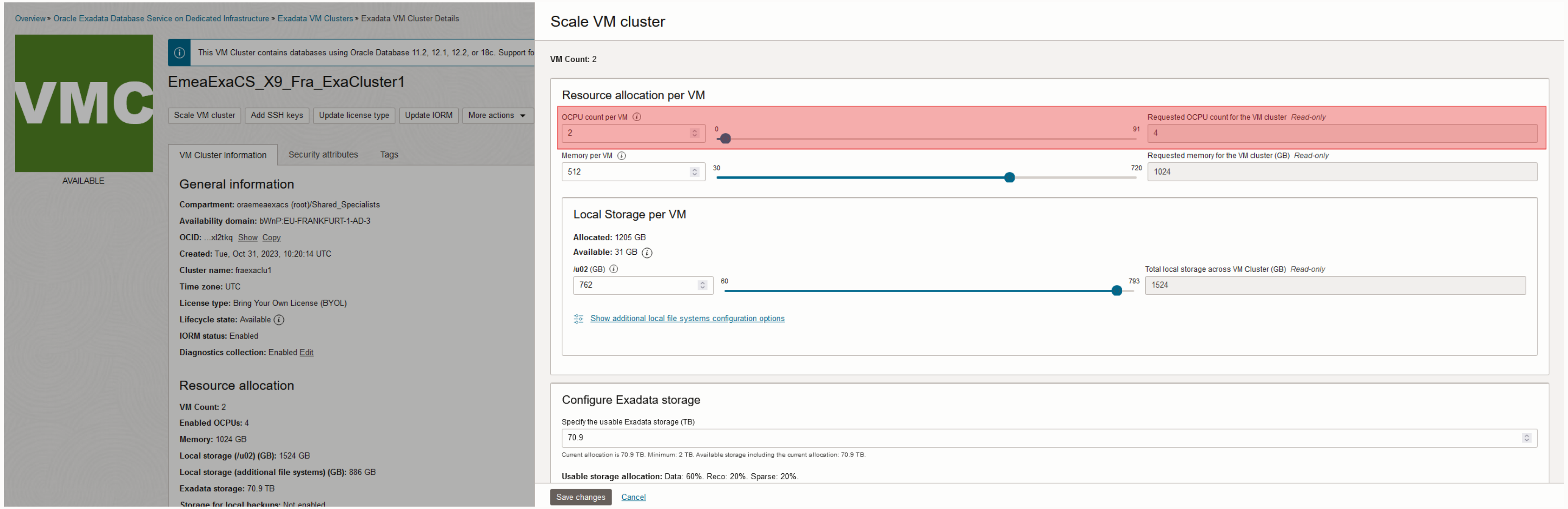


CPU SCALING

- You can create a script using a scripting language of your choice.
- This script to run in the VMs you want to auto scale, or you can orchestrate the process from another server.
- We recommend you set up the script to automatically start and run as a daemon in the background. The script should basically loop indefinitely, measuring the resource (CPU utilization) every period and evaluating the interpreted results of the measurements against the thresholds. If a scaling operation is determined to be required, the script should call the API to scale the VM by the desired amount.
- If you don't want to deal with writing and supporting your own script, Oracle RAC Pack has developed an autoscaling script for Exadata Cloud Service (on OCI or Exadata Cloud@Customer) that you are free to use. You can download it from MOS Note 2719916.1. It takes many of the parameters discussed above and implements some additional features.

Manual OCPUs Scaling Operations

CPU SCALING



UPDATING

Resource allocation

VM Count: 2
Enabled OCPUs: 4
Memory: 1024 GB
Local storage (/u02) (GB): 1524 GB
Local storage (additional file systems) (GB): 886 GB
Exadata storage: 70.9 TB
Storage for local backups: Not enabled
Storage for Exadata sparse snapshots: Enabled

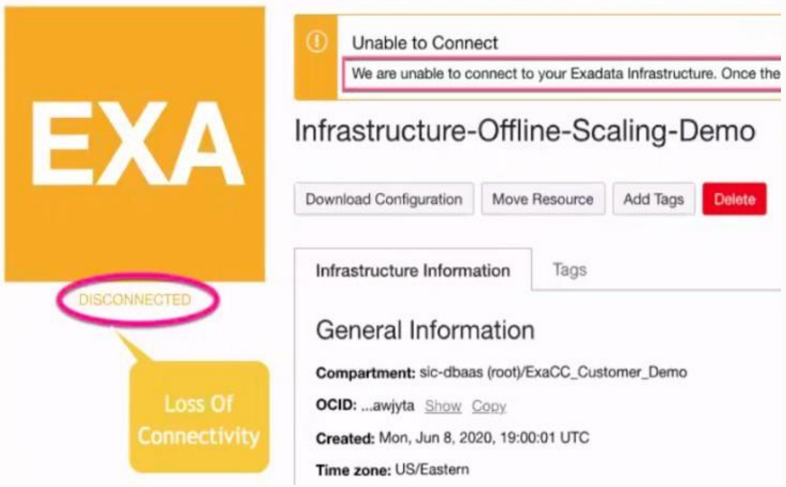
```
$ lscpu -e
CPU NODE SOCKET CORE L1d:L1i:L2:L3 ONLINE
0   0    0      0    0:0:0:0       yes
1   0    0      0    0:0:0:0       yes
2   1    1      1    1:1:1:1       yes
3   1    1      1    1:1:1:1       yes
```

```
SQL > show parameter cpu_count
NAME      TYPE  VALUE
______________________________
cpu_count      integer 4
```

Offline OCPU scaling

- Allows customers to increase or decrease OCPUs allocated to VM Clusters in case of a loss of connectivity to Oracle public cloud

- There are two special commands designed to work in offline mode
  - Must use command-line interface and be connected to ExaC@C Guest VM as root
  - The command once issued applies to all of the nodes the VM Cluster the node belongs
  - The command would work only if the infrastructure is in DISCONNECTED state




```
$ dbaascli cpuscale update –coreCount [--message ]

# dbaascli cpuscale update --coreCount 17 --message EscaladoPorProblemasBatch DBAAS CLI version 20.1.3.0.0
Executing command cpuscale update --coreCount 17 --message EscaladoPorProblemasBatch ** CPU Scale Update **
Input core count (per VM): 17 Message: EscaladoPorProblemasBatch CPU core update request job id:
1608187619926601 Please use 'cpuscale get_status' to know the current job status

dbaascli cpuscale get_status DBAAS CLI version 20.1.3.0.0 Executing command cpuscale get_status ** CPU
Scale Status ** CPU scale job id: 1608187619926601 Status: Success Requested time: Thu Dec 17 07:46:59 CET
2020 Last update time: Thu Dec 17 07:49:25 CET 2020 Message: EscaladoPorProblemasBatch20201217 Scale
request completed successfully

VM Cluster nodes:
# grep -i "core id" /proc/cpuinfo |wc -l 34
```

CPU SCALING

The criteria for performing a scale up or down operation is dependent upon a few different parameters and algorithms that you must set:

The resource you measure:
Typically, you measure CPU utilization within the VM. Since you are scaling CPU, this is the logical resource to measure.

Minimum resource threshold:
A point at which you declare the resource is low. For example, you may decide your CPU utilization should not go below 40%.

Maximum resource threshold:
A point at which you declare the resource is high. For example, you may decide your CPU utilization should not go above 80%.

The frequency of measurement:
How often you take a measurement and evaluate it against your criteria.

Interpretation of the measurement:
This algorithm determines how a measurement will be interpreted and when your script will initiate a scaling operation.

Delay for Scale Down Operations:
To provide optimum service levels, it may be best to not immediately scale down when load drops.
This determines how long should you wait before initiating a scale-down operation.

Automate OCPUs Scaling Operations

About Oracle Dynamic Scaling

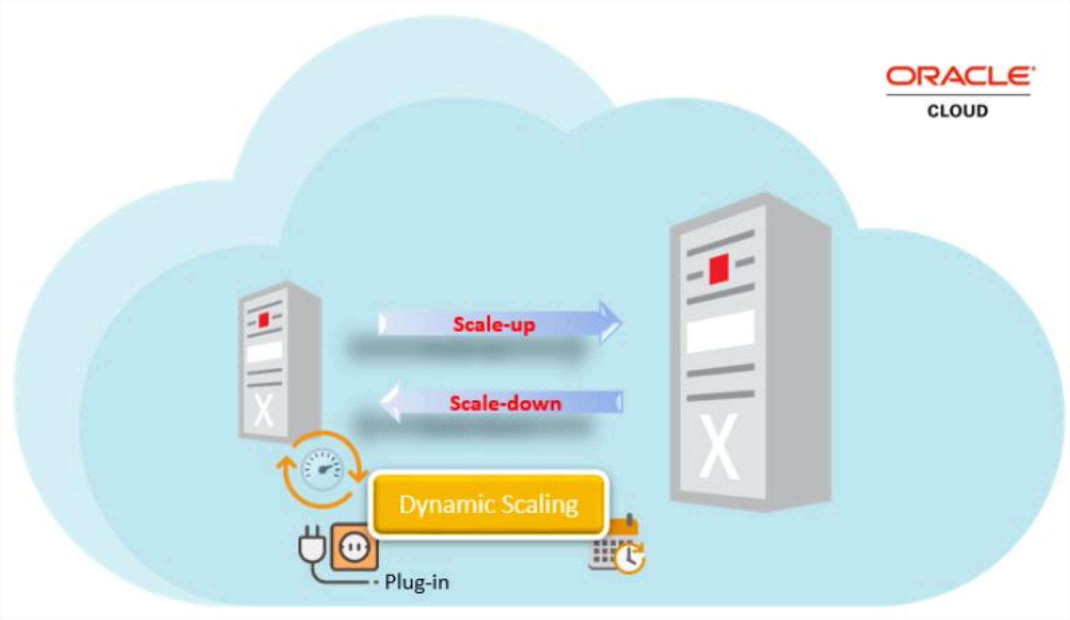
What is Dynamic Scaling?

•A tool to **automate scaling** (up or down) based on:

- **CPU load**
- **Scheduling**

•Ensures resources match workload demands in real-time.

•**Pay only for what you use.**



*(*) first GA release dated **Jan 2021***

Benefits of Oracle Dynamic Scaling

Scale Up/Down Online: Adjust resources without disrupting database operations.

Cost Efficiency: Pay only for the resources actively used.

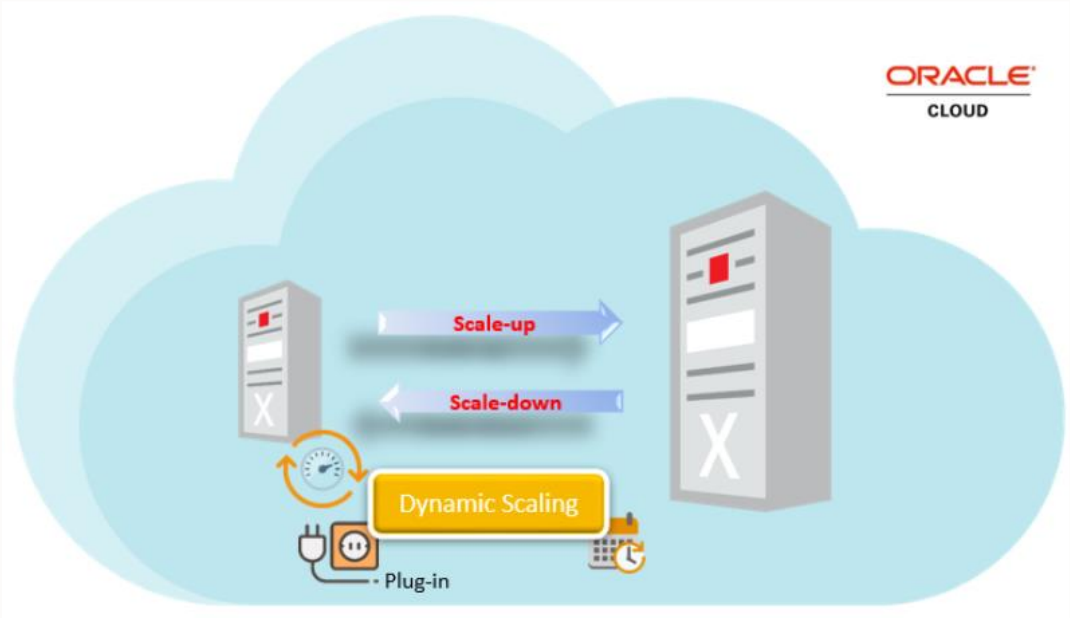
Automated Optimization: System adjusts to workload changes dynamically.

The Problem with Static Sizing

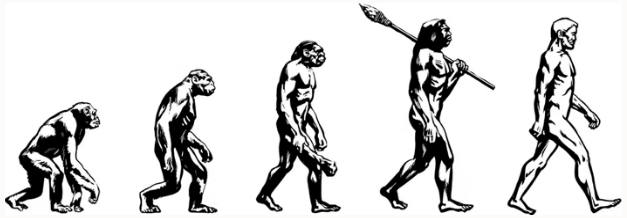
Static VM sizing often leads to:

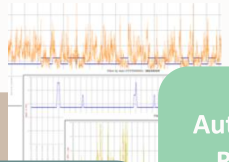
- **Over-provisioning** to avoid frequent resizing.
- **Paying for unused capacity.**

Results in **higher costs** and **inefficient resource utilization.**



Initial Evolution





Autoscaling *1.0.1*

Autoscaling *1.0.2*

- min/max threshold
- dryrun
- restAPI support

Autoscaling *1.0.3*

- Scheduling
- plugin

Autoscaling 2.0.1

- Cluster aware
- Stop, Status, Set
- Debug option
- Proxy options
- New resource model
- 2hr wait time
- Res/inst principals
- Log parameters
- ExaCC X7/X8/X9 support
- ODyS Chart support
- MAX load vs AVG

Autoscaling chart

Autoscaling Remote Plugin

Autoscaling on docker

ODyS Oracle Dynamic Scaling Suite

- ODyS Engine
- ODyS Remote Plugin
- ODyS Chart
- ODyS for Docker

Dynamic Scaling on podman

ODyS Oracle Dynamic Scaling Suite (2.0.1-12)

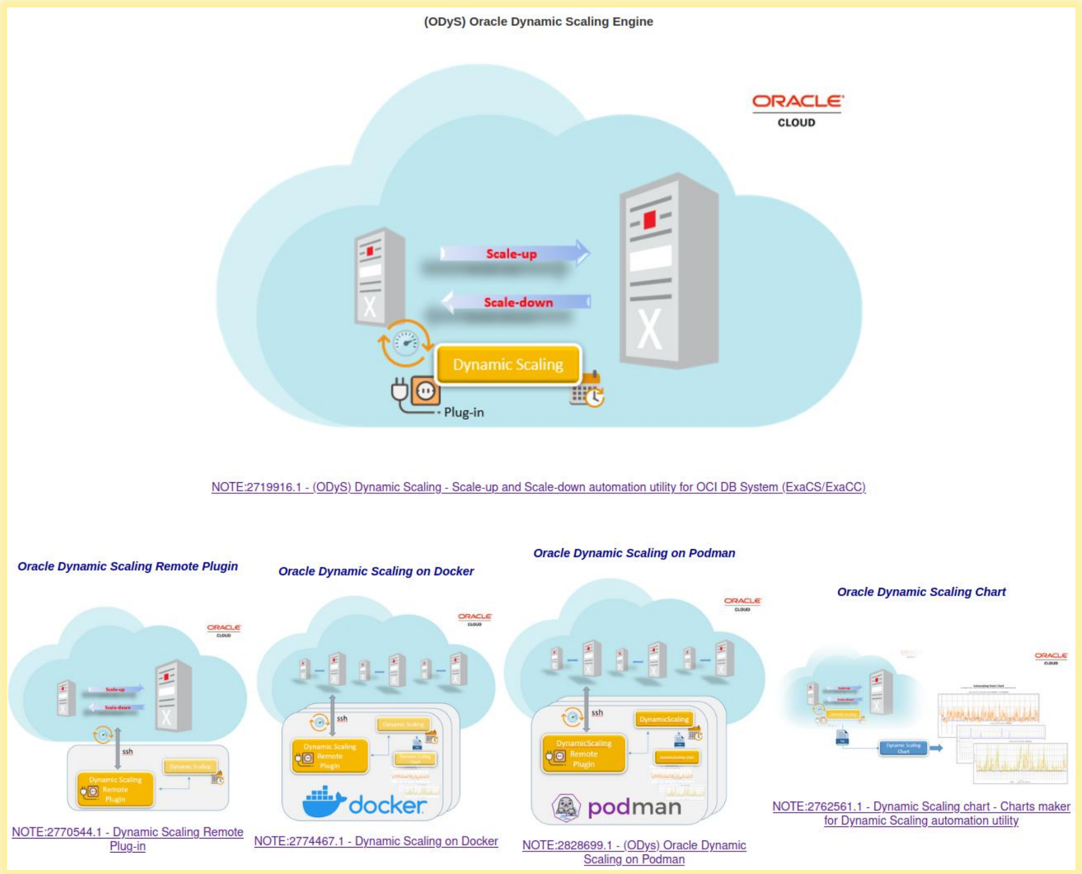
- ScaleUP Warning Notification
- GOV Regions support
- Scale Topic operation (internal)
- Engine/RemotePlugin for nodes down

...

Sept 2020 | Oct 2020 | Nov 2020 | **Jan 2021 First GA release** | Mar 2021 | Apr 2021 | **Dec 2021** | **2022...**

References

- Dynamic Scaling - *Engine*
- Dynamic Scaling - *Remote plugin*
- Dynamic Scaling - *Chart*
- Dynamic Scaling *on Docker*
- Dynamic Scaling *on Podman*
- Dynamic Scaling *on Kubernetes*

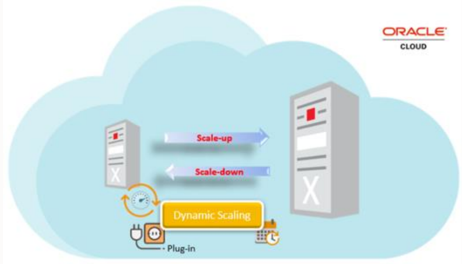


(ODyS) Dynamic Scaling

(ODyS) Dynamic Scaling

Engine

Engine – Key Capabilities



- DynamicScaling can be defined as OS service and it starts at boot time or in case of failure
- DynamicScaling offers “***getocpu***” & “***setocpu***” functions to retrieve or set OCPU “manually” without OCI console access
- DynamicScaling is providing “**check**” function to test OCI connectivity and “**status**” to show if ODyS is running and how with which parameters
- DynamicScaling can retrieve target ExaDB-D/ExaDB-C@C load leveraging on **“external” plugin** bypassing *ODyS load algorithm*
- DynamicScaling can be defined as HA cluster resource following: [(ODyS) How to make Oracle Dynamic Scaling an HA cluster resource (Doc ID 2834931.1)]

Engine –How it works

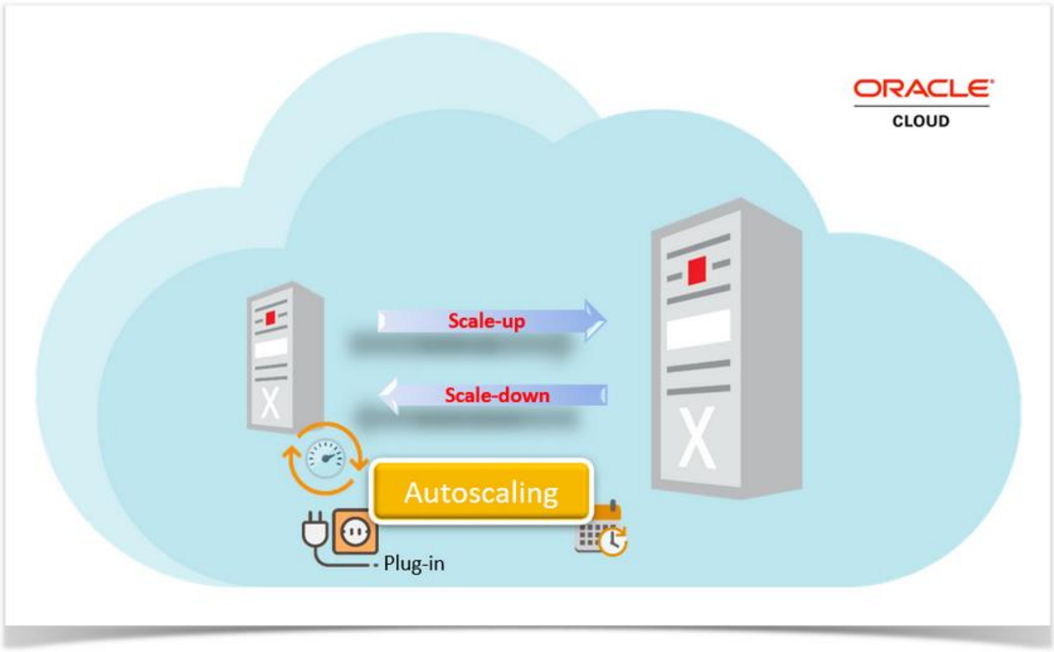
Execution Modes	Standalone or Daemon on one or more ExaDB nodes.
Scale-Up	Triggered if load > --maxthreshold for --interval. Increases OCPU up to --maxocpu.
Scale-Down:	Triggered if load < --minthreshold for --interval. Decreases OCPU down to --minocpu.
Cluster-Wide Load	With ACFS, scales based on AVG or MAX load across nodes.
Grace Period	First scale-down waits 60 minutes after last scale-up.
Scheduling Priority	Scheduling overrides load-based scaling if active. If inactive, scaling is based on load.

Engine – Scale-up Warning Notification Service

Scale-Up Warning Notifications	Triggered when consecutive scale-up operations ≥ sopnum threshold.
Notifications	OCI Notification (using --topic-id). Email (using --mailx, if configured). Indicates potential need to adjust scaling factors or investigate unexpected load.
Reset Conditions	Scale-Down Operations: Resets consecutive scale-up count to zero. Load Below Max Threshold: Resets consecutive scale-up count to zero.
Dry-Run Mode	--dryrun does not disable scale-up warning notifications.
DB System Status Alerts (v2.0.1-15+)	If DB system is in "UPDATING" status for 10 consecutive checks: •Sends OCI Notification or Email to alert subscribers.

Engine – Best Practices

Topic	Guidance
Verify Connectivity	Use the **'check' function** to confirm OCI target connectivity before any actions.
Set OCPU Limits	Ensure **maxocpu** and **minocpu** values are different and have a sufficient margin.
Adjust Timings & Thresholds	Choose **interval** and **CPU thresholds** that match your workload's scaling needs.
Manual Scaling ('setocpu')	- Temporarily resets OCPU values. - Changes are not permanent unless configuration is reset.
Threshold Selection	Use **different thresholds** for scale-out (up) and scale-in (down) based on workload needs.
Weekend Scaling	- DynamicScaling can run at minimum capacity on weekends. - To scale up during scheduled low-capacity periods: - Stop DynamicScaling on all nodes. - Use **'setocpu'** to set the desired OCPU value. - **Note:** Scheduling has priority over load; manual scaling may be overridden.



(ODyS) Dynamic Scaling

Remote Plugin

Remote-Plugin: Key Features

External Execution

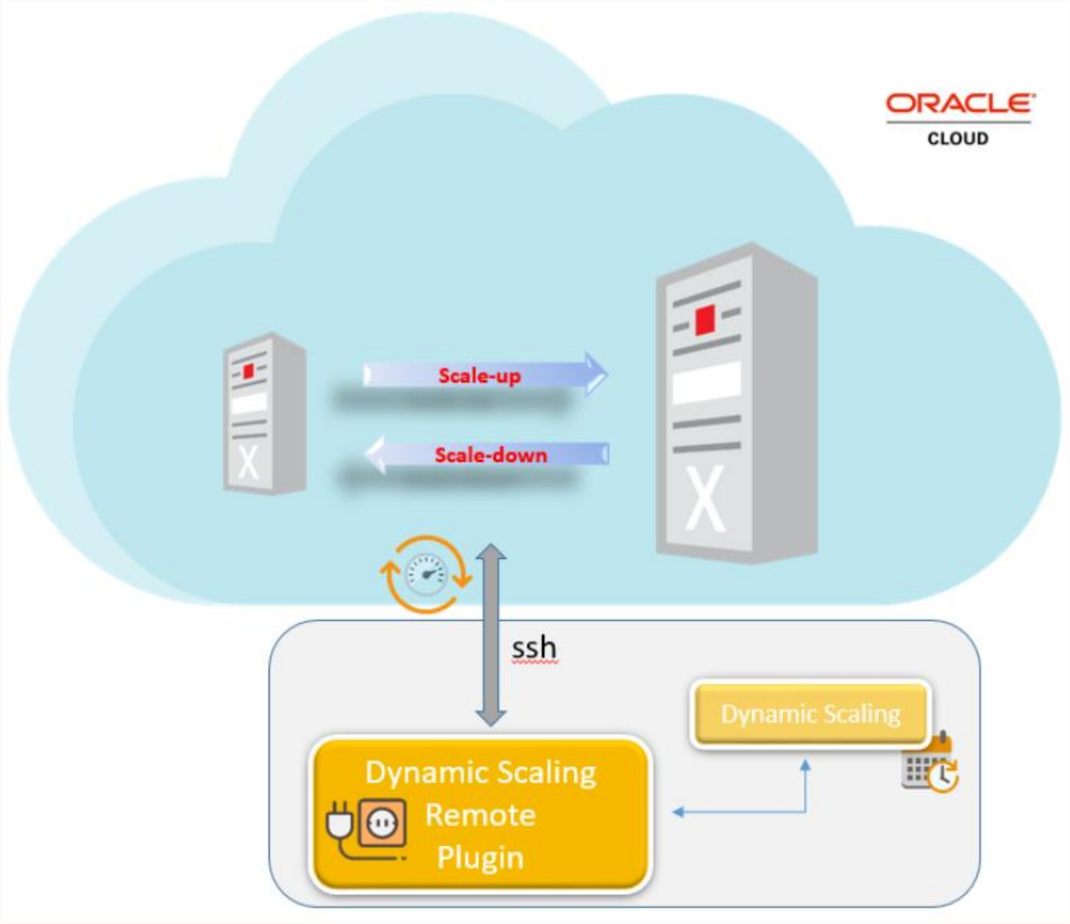
- Runs on a compute host outside ExaDB-D/ExaDB-C@C.
- No RPM installation required on ExaDB systems.

Node Discovery & Data Collection

- Discovers and connects to ExaDB nodes via SSH as the 'opc' user (*not limited to* '*opc*').
- Collects load information from target nodes (or a subset).
- Calculates average or max load and provides it to the DynamicScaling Engine.

Safety Mechanism (v1.0.1-5+)

- If a target node is unreachable, the scale-down operation is skipped.
- Ensures stability and avoids unintended resource reductions.



(ODyS) Dynamic Scaling

Chart

Chart – How is woring?

Key Metrics

- Current OCPU (--cocpu)
- Cluster Load (--cload)
- Node Load (--nload)
- Plugin Load (--pload)

Output Options

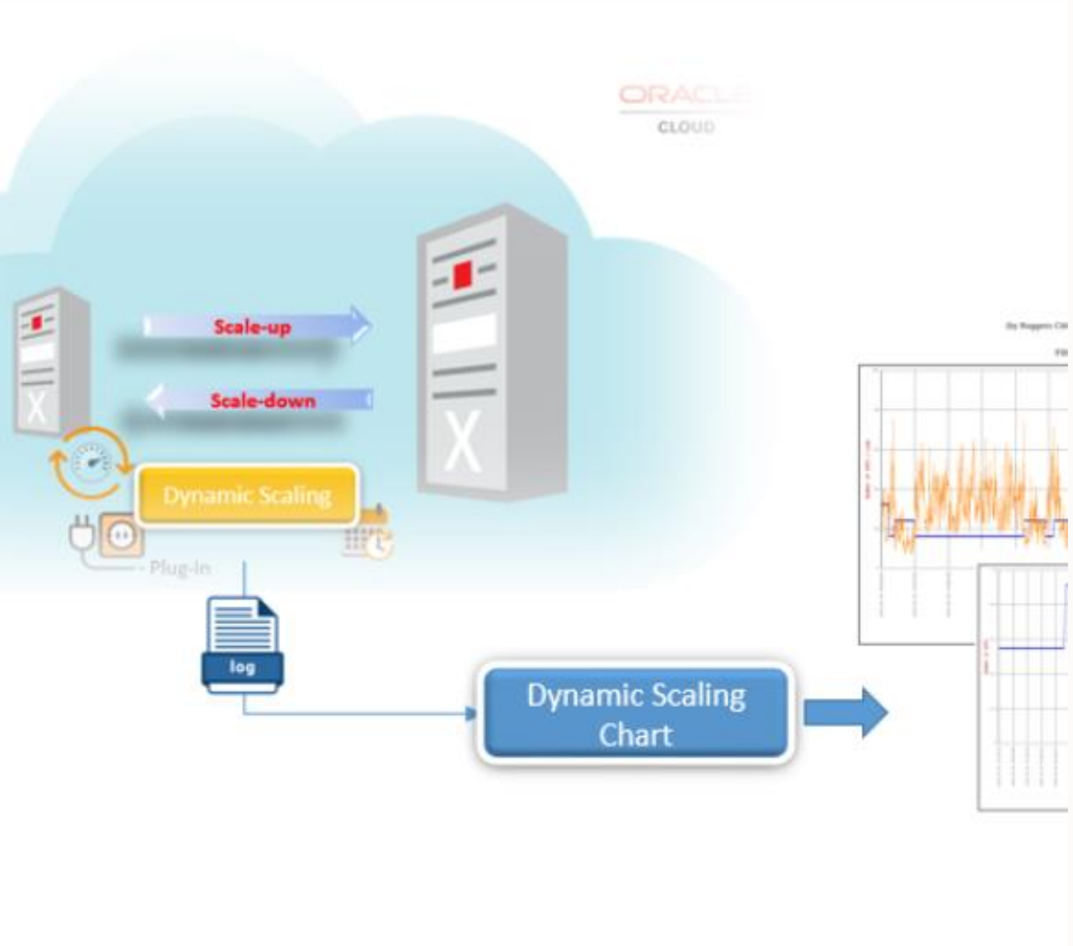
- Default: HTML page with embedded charts and parameters.
- Additional:
  - CSV file (--csv)
  - PNG image (--png)

Filtering

- By Date: --datefilter (e.g., 20210322)
- By Period: --begin/--end

Remote Logs

- Retrieve logs via SSH:
- Example: --log ssh://<username>@remotehost:/<path>



Fleet OCPUs Scaling Operations

(ODyS) Dynamic Scaling

For Docker, Podman

DS4Docker / DS4Podman – How is working

- Using DynamicScaling-plugin you can have DynamicScaling engine running on a remote compute host.
- In the case you need to control multiple ExaDB-D/ExaDB-C@C environments (fleet) you can make DynamicScaling running from docker/podman
- DynamicScaling on docker/podman is an **automatic docker/podman builder** making a docker/podman with *DynamicScaling engine*, *DynamicScaling-plugin* and *DynamicScaling-chart* already configured and running based on input parameters.
- You can have multiple docker/podman controlling multiple ExaCS/CC environments.
- **No Docker/Podman skills are required**

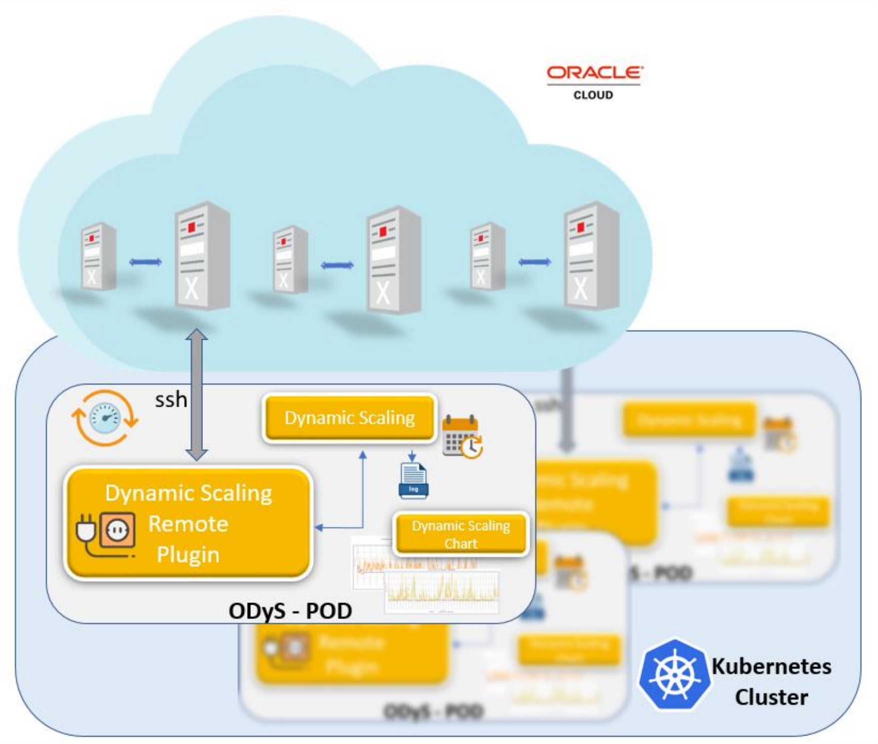
(ODyS) Dynamic Scaling

For Kubernets

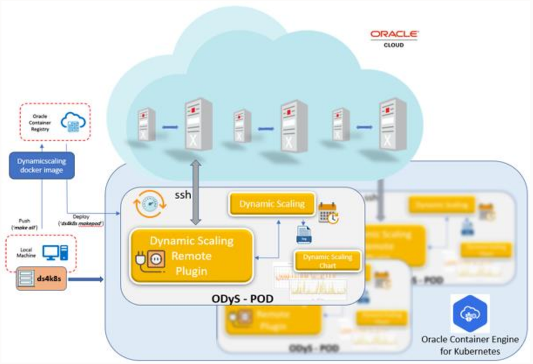
ODyS Kubernetes

- Automated Kubernetes Pod Builder
- Pre-configured with:
  - DynamicScaling
  - EngineDynamicScaling
  - PluginDynamicScaling Chart

Easily manage multiple ExaDB-D & ExaDB-C@C environments with dedicated pods.



Dynamicscaling for kubernets (ds4k8s) needs
- ***kubectl*** - The Kubernetes command-line tool allows you to run commands against Kubernetes clusters
- Target Kubernetes clusters such ***OKE*** (Oracle Container Engine for Kubernetes). Dynamicscaling for kubernets works and it has been tested on ***Minikube*** too
- ***Dynamicscaling image for Kubernetes***



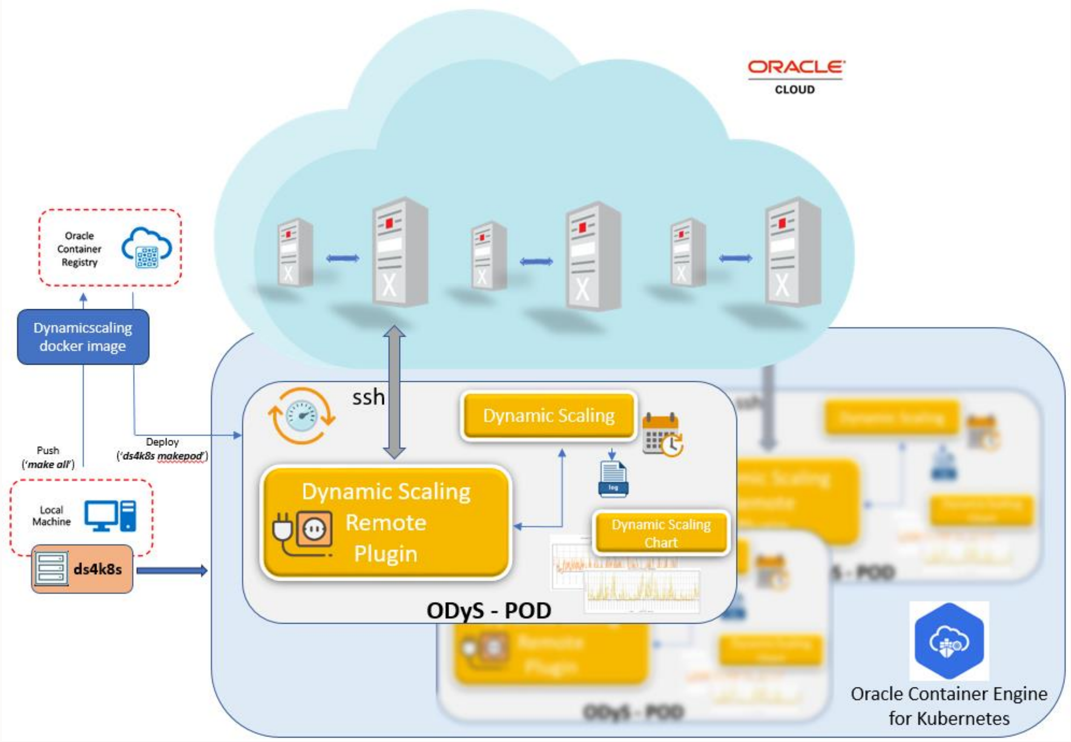
DS4k8s- Verbs

- Check OCI target connectivity:
  - ➢ check
- Check target load:
  - ➢ checkload
- Check notification:
  - ➢ checknotification
- Delete dynamycscaling deployment:
  - ➢ delete
- List dynamycscaling deployment:
  - ➢ list
- Make dynamycscaling deployment yaml file:
  - ➢ makeyaml
- Make dynamycscaling deployment:
  - ➢ makedep
- Make dynamycscaling secrets:
  - ➢ makesecrets
- Make dynamycscaling deployment charts:
  - ➢ chart
- Conecting dynamycscaling deployment pod:
  - ➢ connect
- Get dynamycscaling deployment pod log:
  - ➢ getlog
- Get current target OCPU:
  - ➢ getocpu
- Set current target OCPU:
  - ➢ setocpu
- Show dynamycscaling deployment pod configuration:
  - ➢ showconf
- Show dynamycscaling deployment pod log:
  - ➢ showlog
- Show dynamycscaling deployment pod status:
  - ➢ status
- Start dynamycscaling deployment:
  - ➢ start
- Stop dynamycscaling deployment:
  - ➢ stop

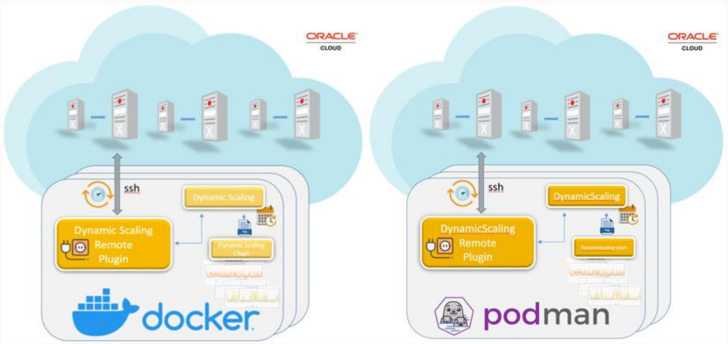
DS4Kubernetes – How is working

DS4K8s Setup in short

1. 'cd /opt/ds4k8s/build_docker_image; **make all'**
2. '**ds4k8s check** <parameters>'
3. '**ds4k8s makedep** <parameters>'



DS4Docker/DS4Podman - Verbs



- Make DynamicScaling Image/Container:
  - make
- Check OCI target connectivity:
  - Check
- Check OCI target load:
  - checkload
- Connect DynamicScaling Container:
  - connect
- Delete DynamicScaling Container:
  - delete
- List DynamicScaling Containers:
  - list
- Restart DynamicScaling Container:
  - restart
- Start DynamicScaling Container:
  - start
- Stop DynamicScaling Container:
  - stop
- Get target OCPU:
  - getocpu
- Set OCPU to target
  - setocpu
- Make DynamicScaling Container command:
  - makecmd
- Show DynamicScaling Container configuration:
  - showconf
- Show Dynamicscaling Container RPMs version:
  - showvers
- Change DynamicScaling Container restart policy:
  - modify
- Update DynamicScaling Container RPM:
  - update
- Make DynamicScaling charts:
  - chart
- Cleanup DynamicScaling docker image and all containers:
  - cleanup

(ODyS) Dynamic Scaling

Considerations & Best Practices

ODyS Considerations & Best Practices

- EXACS required OCI IAM privileges are CLOUD_VM_CLUSTER_UPDATE and CLOUD_EXADATA_INFRASTRUCTURE_UPDATE for which required policy definition could be.

```
Allow group <groupname> to use cloud-exadata-infrastructures IN TENANCY
Allow group <groupname> to use cloud-vmclusters IN TENANCY
```

- EXAC@C required OCI IAM privileges are VM_CLUSTER_UPDATE and EXADATA_INFRASTRUCTURE_UPDATE for which required policy definition could be.

```
Allow group <groupname> to use exadata-infrastructures IN TENANCY
Allow group <groupname> to use vmclusters IN TENANCY
```

- Once dynamicscaling is firing a scale up/down request, DB System needs about 5 minutes to complete the operation.

- Consider disabling dynamicscaling during your DB host lifesycle operations like patching, upgrade etc.

ODyS Considerations & Best Practices

Optimize Your Scaling with DynamicScaling

- Run with minimal capacity on weekends to reduce costs.
- Before scaling up manually, stop DynamicScaling on all nodes.
- Scheduled scaling takes priority over manual adjustments.
- Key Resources:
  - Possible Configurations – [MOS Note 2834931.1]
  - Load Imbalance? – Check ODyS Remote Plug-in [MOS Note 2770544.1]

Dynamic CPU Count

Dynamic CPU Count

Overview

- *Automation Utility: Dynamically adjusts cpu_count in Oracle Database.*
- *Works with Instance Caging and Resource Manager to optimize CPU allocation across multiple instances.*

Key Features

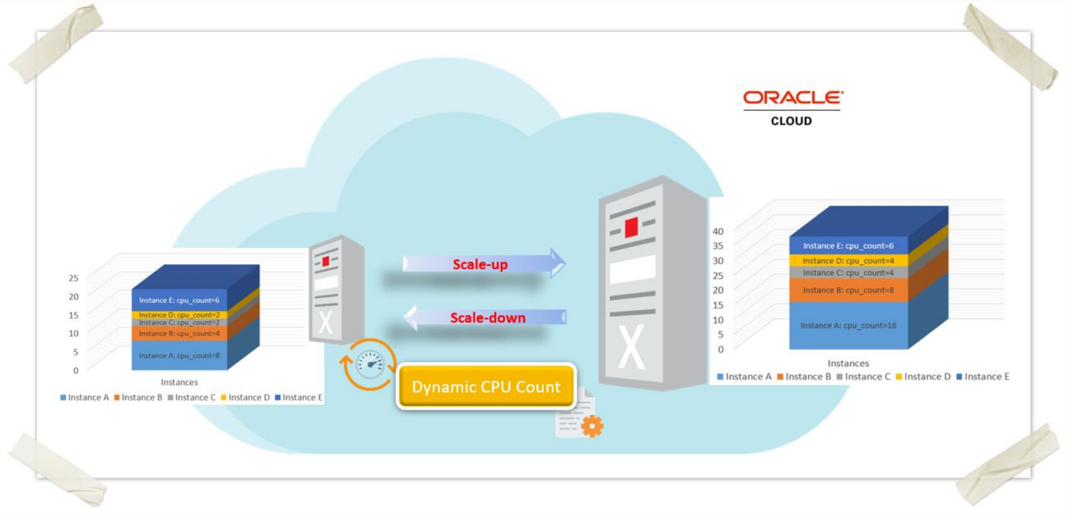
Sets cpu_count based on:

  - *configuration file.*
  - *percentage of available host OCPUs.*

Integration

- *Designed to work alongside Oracle Dynamic Scaling for seamless CPU management.*

Version 2.0.1-1+: Supports specifying a fixed number for cpu_count.
Version 3.0.1+: Adds support for pluggable databases (PDBs).



(ODyS) Dynamic Scaling

References

ODyS References

(ODyS) Oracle Dynamic Scaling Suite Main Index Page (Doc ID 2774779.1)
(ODyS) Scale-up and Scale-down automation utility for OCI DB System (ExaCS/ExaC@C) - (Doc ID 2719916.1)
(ODyS) Oracle Dynamic Scaling - Remote Plug-in - (Doc ID 2770544.1)
(ODyS) Oracle Dynamic Scaling - Charts maker for Dynamic Scaling engine utility - (Doc ID 2762561.1)
(ODyS) Oracle Dynamic Scaling on Docker - (Doc ID 2774467.1)
(ODyS) Oracle Dynamic Scaling on Podman - (Doc ID 2828699.1)
(ODyS) Oracle Dynamic Scaling on Kubernetes (Doc ID 2940064.1)
(ODyS) How to make Oracle Dynamic Scaling an HA cluster resource - (Doc ID 2834931.1)
HowTo configure oci-cli with Instance/Resource Principals - (Doc ID 2763990.1)
(ODCC) Dynamic CPU Count - automation utility to change cpu_count dynamically (Doc ID 2915837.1)
(OCI) FixOCPU - automation utility to pin OCPU for OCI DB System (ExaDB-D/ExaDB-C@C) (Doc ID 2913886.1)

(ODyS) Dynamic Scaling

Conclusions

ODyS Conclusions

Reduce Your Cloud TCO with Autoscaling

- Stop paying for excess capacity – Scale only when needed.
- Autoscaling matches workload demands with Exadata capacity for true pay-for-use pricing.
- Easy automation with published APIs and script-driven scaling.
- Get started today:
  - Use your favorite scripting language (see article).
  - Download a fully tested solution from MyOracle Support:

MOS Note 2719916.1 – Oracle Dynamic Scaling (ODyS) for ExaCS/ExaC@C.

ORACLE